Quantitative Least-Term Transfers and Character-Block Jet Certificates

A scoped study of maximal Seshadri constants on polarized surfaces

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Abstract

The source manuscript claims maximal multipoint Seshadri constants on every polarized smooth complex projective surface for all sufficiently large numbers of very general points. We study its actual finite jet mechanisms without presenting that global claim as independently certified. Positive anisotropic weights give a computable finite-search gap and a convergent coefficient majorant. A tail-annihilating left inverse can raise the effective decay order of the jet defect, subject to an exact disjoint-image criterion; an analytic two-column example removes arbitrarily large first-order coefficients. On the source's finite torus subgroup, normalized Euler jets split unitarily into integer moment blocks. We derive blockwise conditioning and leakage budgets while retaining the nonunitary costs of coordinates and frames. The report maps the entire cached source, proves the local statements, and separates finite certificates from the all-order geometry, countable-open argument, and nefness conclusion. Contribution assessment: C3/5, the source-specific cancellation and character-block error bounds give a notable local certification improvement, not a new maximal-Seshadri theorem.

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1 Scope, status, and the geometric target

Let S be a smooth integral complex projective surface, L an ample line bundle, $H = L^2 > 0$, and $\boldsymbol{p} = (p_1, \dots, p_r)$ distinct points. With $\pi : Y \to S$ their blow-up and E_i its exceptional curves, set

$$\varepsilon(S, L; \boldsymbol{p}) = \inf_C \frac{L \cdot C}{\sum_i \operatorname{mult}_{p_i} C} = \sup\{a \geq 0 : \pi^* L - a \sum_i E_i \text{ is nef}\}. \tag{1}$$

The infimum uses integral curves meeting the tuple. The universal upper bound is $\sqrt{H/r}$. The source [1] asserts that some $r_0(S, L)$ gives equality at very general tuples for every integer $r \geq r_0$. Here "very general" means outside a countable union of proper Zariski-closed subsets of the distinct-point configuration space U_r. This report does not upgrade that source assertion to an accepted or independently verified solution of the qualitative Nagata–Biran conjecture. Applicability L3 concerns a finite transfer mechanism, not proof validity; the separate C3 score concerns only the demonstrated local increment.

1.1 Whole-source dependency map

The cached source has ten TeX files, a bibliography, and a README. All 1,662 lines of these twelve files were literally displayed in bounded chunks for this report. Reading is not certification of every cited theorem. The following map includes the geometric modules, not just the rank lemma.

Module	Role and retained boundary
Introduction	Precise all-large-r claim, comparison with packing and bounded-multiplicity interpolation; the companion plane claim is not a premise.
Jets	Full column rank; local units and coordinate changes preserve orders; least-term basis, a small weight perturbation, and analytic dilation.
Node	An integral divisor in $\lvert dL\rvert$ with one ordinary node supplies two local branches. Removing its powers controls initial exponents, not the entire Taylor support.
Compression	Diagonal exponent slices of a quadrilateral become vertical slices of a triangle by a finite change of basis and an analytic limit near $v = 1$.
Lattice	Triangle multiplicity estimate, exact character projections, index-r subgroup, and a continuously tuned direction fixed before degree and multiplicity.
Nefness	Algebraic openness, countably many tests, positive-square Riemann–Roch, unequal curve multiplicities, upper bound, and characteristic-zero descent.
Ancillary	Two figures illustrate containing regions; main/preamble assemble the manuscript; bibliography locates imported results.

In source notation, choose d, H, T with $t > T > 1$ and $(1+t)^2 > d^2 Ht$. Set $c = 1/d$, $a = dHt/(1+t)$, $b = dH/(1+t)$. The initial exponents of $H^0(S, L^{\otimes k})$ lie in

$$kQ_t, \quad Q_t = \operatorname{conv}\{(0,0), (a,0), (c,c), (0,b)\}.$$

If a section has nodal-divisor factor τ^j, then $k - jd \geq 0$ and its initial exponent is (j, j) plus that of the residual section. For a residual section, the two branch intersection orders satisfy $v(1 + 1/t) \leq kdH$. This is the source's geometric input to the polygon; our analytic majorants do not prove that input.

Under $(x, z) = (u, v/u)$, the exponent coordinate is $q = \alpha - \beta$. Consecutive lattice points in a slice are replaced by $u^q v^{\beta_q^-}(v-1)^h$. Substituting $v = 1 + \sigma y$ and dividing by σ^h produces $u^q(1+\sigma y)^{\beta_q^-} y^h$. Its limit is $u^q y^h$ with exponent in

$$kP_t, \qquad P_t = \operatorname{conv}\{(-b, 0), (a, 0), (0, c)\}, \qquad \operatorname{area}(P_t) = H/2.$$

The coordinate map has Jacobian $\sigma/u \neq 0$ at the chosen torus points. It is not an isometry of jet norms. The source already proves this qualitative compression; it is not claimed here as new row reduction.

1.2 Literature at the stated date

Syzdek–Szemberg's Conjecture 4.3 [2] formulates the all-large-r maximality target. Their asymptotic comparison does not assert exact maximality for every large integer. Alexander–Hirschowitz [3] fix a multiplicity bound before taking the degree sufficiently large; this order of quantifiers is not the source's simultaneous all-order test. Dionne–Roth [4] treat specified regions and values on $\mathbb{P}^1 \times \mathbb{P}^1$, not arbitrary polarized surfaces. The September 22, 2026 preprint of Laface–Ugaglia [5], Theorems 1–2, claims the ten-point value $1/\sqrt{5}$ for $\mathcal{O}(1, 1)$ and the one-point value $2\sqrt{5}$ for a specified ample divisor on a nine-point plane blow-up. These are recent preprint claims, not global inputs verified by this report. Public primary statements were checked afresh; no confidential title or proof passage was used in an online query. The October 1 preprint of Malara–Merta–Szpond–Zieliński [6] claims a related family for every odd $n \geq 5$, including the $2n$-point value $\sqrt{(n-4)/n}$ for $\mathcal{O}(n-4, 1)$. This update was checked at the primary abstract only, not at its proof; it is not invoked below.

2 The exact finite observation problem

Fix integers $k, m, r > 0$ for the rest of each finite construction. For a finite-dimensional holomorphic germ space F of dimension n, a basis $f_1, \ldots, f_n$, and distinct points $q_1, \ldots, q_r$ in a common coordinate domain, use Taylor-normalized rows

$$J(F)_{(j,\nu),i} = \frac{\partial^\nu f_i(q_j)}{\nu!}, \qquad \nu = (\nu_1, \nu_2) \in \mathbb{N}^2, \quad |\nu| < m. \tag{2}$$

There are $h = \binom{m+1}{2}$ rows per point and rh total rows. Injectivity is rank $J = n$, requiring $n \leq rh$. It is not surjectivity onto arbitrary jets. A left inverse reconstructs compatible data only; it does not turn an overdetermined residual equation into an exact solvability statement.

The private original finite-moment treatment [7], at `eq:moments`, explicitly requires full column rank and a specified left inverse for a redundant bank. Its polynomial construction distinguishes positive integration weights from a positive discrete Gram matrix, which also needs injectivity. The original local-moment proof [8] keeps an absolute Taylor remainder and its factorial normalization, and distinguishes output-centered from source-centered coefficient derivatives. Here the observations are exact holomorphic jets, not an integral operator: we use the finite response and remainder discipline, not an unjustified identity between the source's algebraic geometry and a PDE model.

The original perturbation proof [9] gives the familiar $\gamma - \eta$ singular-value margin and warns that a rectangular left inverse cannot certify an arbitrary nonlinear root. Our use of a left inverse

below is only to prove a trivial kernel of a linear map. None of these classical matrix facts alone earns contribution credit.

3 Positive-weight gaps and explicit analytic tails

Put $w(\alpha, \beta) = \alpha + \tau\beta$ for a fixed real $\tau > 0$. After the source's finite basis elimination and small tie-breaking change of weight, write

$$f_i(x, z) = x^{e_{i1}} z^{e_{i2}} + \sum_{e \neq e_i} c_{ie} x^{e_1} z^{e_2}, \qquad c_{ie} \neq 0 \Longrightarrow w(e) > \lambda_i := w(e_i). \tag{3}$$

The distinct e_i are unchanged by that weight perturbation; the source proves its existence for this finite space, not uniformly in all k. For $0 < s < 1$ define

$$g_{i,s}(X, Z) = s^{-\lambda_i} f_i(sX, s^\tau Z).$$

Real positive s makes real exponents unambiguous. These functions are holomorphic in (X, Z); no holomorphic dependence on s is asserted.

Lemma 3.1 (A genuine positive gap)**.** *For any $\lambda \geq 0$ and any subset $A \subset \{e \in \mathbb{N}^2 : w(e) > \lambda\}$, the positive differences $w(e) - \lambda$ have a positive common lower bound. One valid bound is*

$$\delta = \min\big(\{1\} \cup \{w(e) - \lambda : e \in A,\ \lambda < w(e) < \lambda + 1\}\big) > 0. \tag{4}$$

The displayed search is finite. If $\tau = p/q > 0$ with positive integers p, q and $\lambda = w(e_i)$, then $\delta \geq 1/q$ may be used instead.

Proof. The inequality $w(e) < \lambda + 1$ bounds $e_1 < \lambda + 1$ and $e_2 < (\lambda + 1)/\tau$. There are only finitely many nonnegative integer pairs under those bounds, and all differences in the finite set are strictly positive. Outside it the difference is at least one. For rational τ each difference is an integer multiple of $1/q$. An empty set A causes no problem and permits any positive decay exponent. □

Positivity of both weights is essential. The statement does not follow from "irrational weights have no ties" alone. Exact support information can improve the gap; replacing A by all eligible lattice exponents gives a conservative finite certificate. To compute a certified positive lower bound with irrational input weights, exact comparisons or rigorous interval separations are needed, not rounded comparisons near a tie.

Theorem 3.2 (Tail bound in the source's dilation)**.** *Assume f_i is holomorphic on a neighborhood of the closed polydisc $|x| \leq R_x$, $|z| \leq R_z$ and $|f_i| \leq B_i$ there. Choose $P_x, P_z > 0$, $0 < s_0 < 1$, with $q_x = s_0 P_x / R_x < 1$, $q_z = s_0^\tau P_z / R_z < 1$. Let $\mathcal{A}$ be a finite set of tail exponents to be removed. Suppose every remaining nonzero tail coefficient in column i has $w(e) - \lambda_i \geq \Delta_i > 0$. Define*

$$H_i = \frac{B_i s_0^{-\lambda_i}}{(1 - q_x)(1 - q_z)}, \tag{5}$$

$$R_{i,s} = g_{i,s} - X^{e_{i1}} Z^{e_{i2}} - \sum_{e \in \mathcal{A} \setminus \{e_i\}} c_{ie} s^{w(e) - \lambda_i} X^{e_1} Z^{e_2}, \tag{6}$$

where c_{ie} denotes only the tail coefficient, zero at e_i. For $0 < s \leq s_0$, on $|X| \leq P_x$, $|Z| \leq P_z$,

$$|R_{i,s}| \leq H_i (s/s_0)^{\Delta_i}. \tag{7}$$

If $|q_{jx}| + \rho_x \leq P_x$ and $|q_{jz}| + \rho_z \leq P_z$ for positive margins ρ_x, ρ_z, its normalized jet error obeys

$$\left| \frac{\partial^\nu R_{i,s}(q_j)}{\nu!} \right| \leq \rho_x^{-\nu_1} \rho_z^{-\nu_2} H_i (s/s_0)^{\Delta_i}. \tag{8}$$

Proof. Cauchy's coefficient inequality is $|c_{ie}| \le B_i R_x^{-e_1} R_z^{-e_2}$ for tail coefficients. At s_0, the absolute sum of their scaled monomials on the polydisc is at most $B_i s_0^{-\lambda_i} \sum_{a,b\ge 0} q_x^a q_z^b = H_i$. For each retained term its factor at s divided by that at s_0 is $(s/s_0)^{w(e)-\lambda_i} \le (s/s_0)^{\Delta_i}$. Absolute convergence justifies the termwise bound and subtraction of the finite sum. Cauchy's derivative formula on the polydisc of radii (ρ_x, ρ_z) about q_j proves the normalized jet estimate. □

For $\mathcal{A} = \varnothing$, Lemma 3.1 supplies Δ_i. For a desired higher exponent Δ, all possibly nonzero terms with $0 < w(e) - \lambda_i < \Delta$ form a finite set. Their removal is legitimate in an estimate only if they are evaluated explicitly or annihilated by the actual observation functional; they cannot simply be discarded. The constants require a true holomorphic bound B_i, not a sampled maximum.

4 Tail-annihilating relative certificates

Let v_e be the Taylor jet column of $X^{e_1} Z^{e_2}$ at the fixed tuple, and $J_0 = [v_{e_1} \ \cdots \ v_{e_n}]$. For a finite set $\mathcal{A}$ of unwanted tail directions put $B = [v_e]_{e\in\mathcal{A}}$; dependent columns in B are allowed. The following exact criterion is useful before making any numerical estimate.

Proposition 4.1 (Feasibility of cancellation)**.** *There is a matrix $L : \mathbb{C}^{rh} \to \mathbb{C}^n$ with*

$$LJ_0 = I_n, \qquad LB = 0 \tag{9}$$

if and only if J_0 has full column rank and $\operatorname{im} J_0 \cap \operatorname{im} B = \{0\}$*. Equivalently,* $\operatorname{rank}[J_0 \ B] = n + \operatorname{rank} B$*. These are exact rank conditions.*

Proof. If $J_0 c = Bd$, applying L gives $c = 0$ and the common vector is zero; also $LJ_0 = I$ implies injectivity. Conversely define $L(J_0 c + Bd) = c$ on $\operatorname{im} J_0 + \operatorname{im} B$. The zero intersection and injectivity make this well-defined. Extend linearly to the ambient finite space. The dimension formula gives the rank equivalence. All of this can be done by exact elimination over a field containing the point coordinates. □

This is an elementary quotient-space fact, not a new linear algebra theorem. Its useful consequence here is a higher exponent in the *source's anisotropic scale*. Cancellation is impossible, for example, if a selected tail direction is a nonzero initial column. It also cannot annihilate nonzero extra directions when J_0 is square and invertible. Redundant jet observations are a resource with a precise limitation.

Write $a_{(j,\nu)} = \rho_x^{-\nu_1} \rho_z^{-\nu_2}$ and $h_i(s) = H_i (s/s_0)^{\Delta_i}$. These are positive real quantities. For positive coefficient weights d_i use $\|c\|_{\infty,d} = \max_i |c_i|/d_i$.

Theorem 4.2 (Effective gap after cancellation)**.** *Under Theorem 3.2 and (9), let J_s be the jet matrix of $g_{1,s}, \dots, g_{n,s}$. Put*

$$\kappa(s) = \max_{1\le p\le n} \frac{(|L|a)_p}{d_p} \sum_{i=1}^{n} h_i(s) d_i. \tag{10}$$

If $\kappa(s) < 1$, then J_s is injective and has the left inverse

$$L_s = (LJ_s)^{-1} L, \qquad \left\| (LJ_s)^{-1} \right\|_{\infty,d} \le \frac{1}{1-\kappa(s)}. \tag{11}$$

For any observation noise e and compatible exact coefficients c, $\|L_s(J_s c + e) - c\|_{\infty,d} \le \|Le\|_{\infty,d}/(1-\kappa(s))$. In particular, if $\Delta_i \ge \Delta$ for all i and $C = \max_p (|L|a)_p/d_p \sum_i H_i d_i$, then any $s \le s_0$ satisfying $C(s/s_0)^{\Delta} < 1$ works.

Proof. Expand the columns as in (6). Applying L kills every explicit v_e tail term and sends the initial part to I. For the remainder matrix E_s, (8) gives the entrywise bound $|LE_s| \le (|L|a)h(s)^T$. Its weighted row sums are bounded by (10). The Neumann series for $I + LE_s = LJ_s$ converges in the induced weighted norm. An invertible LJ_s forces $\ker J_s = 0$, and gives (11). Applying this left inverse to the noisy data proves the last estimate. No claim of arbitrary-data solvability is used. □

4.1 Imperfect algebra and numerical input

The certificate need not pretend that a computed L is exact. Let $t_{ie}(s) = c_{ie}s^{w(e)-\lambda_i}$ denote the explicit tail coefficients, zero when absent. If the implemented observation matrix has an entrywise error bound $\Omega \ge 0$, a sufficient majorant for its preconditioned deviation from identity is

$$K(s) = |I - LJ_0| + \sum_{e\in\mathcal{A}} |Lv_e|\,(|t_{1e}(s)|, \dots, |t_{ne}(s)|) + (|L|a)h(s)^T + |L|\Omega. \tag{12}$$

Every absolute value here is entrywise. If $K(s)d \le \kappa d$ with $d > 0$ and $\kappa < 1$, the same argument applies to that implemented matrix. To certify the true matrix, use a majorant covering its error relative to the matrix actually tested. A small residual without such a covering bound gives no exact rank certificate. Formula (12) can exploit the actual small vectors Lv_e without exact cancellation; replacing everything by $\|L\|\,\|J_s - J_0\|$ loses this information.

Example 4.3 (An explicit two-column gain)**.** Take $\tau = 2$, $q = (1, 1)$, $m = 3$, and germs

$$f_1 = 1 + Ax + B\frac{x^2}{1-x}, \qquad f_2 = z + Cxz + D\frac{x^2 z}{1-x}, \quad A, C \in \mathbb{C}, \quad B, D \ge 0. \tag{13}$$

They are holomorphic for $|x| < 1$ with no restriction on z. Their initial exponents are $(0, 0), (0, 1)$, of weights $0, 2$. After dilation the two columns are the jets of

$$1 + AsX + B\frac{s^2X^2}{1-sX}, \qquad Z\left(1 + CsX + D\frac{s^2X^2}{1-sX}\right).$$

Define functionals, evaluated at $(1, 1)$,

$$\ell_1 = (1 - \partial_Z)(1 - \partial_X), \qquad \ell_2 = \partial_Z(1 - \partial_X).$$

All their derivatives have total order at most two and are available in the $m = 3$ jet bank. On the initial pair their matrix is I_2; they both kill the tail directions X, XZ. Using $\sum_{j\ge0}(j+1)s^j = (1-s)^{-2}$, one obtains exactly

$$LJ_s = \operatorname{diag}\left(1 - \frac{Bs^2}{(1-s)^2}, 1 - \frac{Ds^2}{(1-s)^2}\right). \tag{14}$$

For $s = 1/4$, $B = 1$, $D = 2$ the relative defect is $2/9 < 1$, independently of A, C. A raw entrywise error bound contains the terms $|A|s, |C|s$ and cannot stay small as these grow. The effective order has risen from one to two, while the germ space and the actual jet order are unchanged. This is a strict gain in a sufficient certification budget, not an assertion that the true rank was previously deficient. If only sampled values of A, C were supplied, exact cancellation would have to be replaced by (12).

The example gives analytic, not merely polynomial, tails and an exact closed-form check. More aggressive cancellation can be attempted by adding finite weight shells, but must pass Proposition 4.1 each time. Its potentially larger left-inverse norm and increased jet order can outweigh the improved exponent.

5 Character blocks of the source's torus test

Let $M \subset \mathbb{Z}^2$ be a sublattice of index r and $G = \{g \in (\mathbb{C}^*)^2 : \chi_u(g) = 1 \text{ for all } u \in M\}$, where $\chi_e(x, z) = x^{e_1} z^{e_2}$. Then G has r points. Each coordinate of any g is a root of unity, hence has modulus one: $r\mathbb{Z}^2 \subset M$ suffices. Let $\Omega \subset \mathbb{Z}^2$ be a finite set of Laurent exponents, $N = |\Omega|$, and Ω_η its coset slices for $\eta \in \mathbb{Z}^2/M$, of sizes n_η.

Put $D_X = X\partial_X$, $D_Z = Z\partial_Z$. Fix arbitrary positive row weights d_ν, $|\nu| < m$. The normalized Euler observation matrix is

$$\mathcal{J}_{(g,\nu),e} = r^{-1/2} d_\nu e_1^{\nu_1} e_2^{\nu_2} \chi_e(g). \tag{15}$$

The convention $0^0 = 1$ is used; negative integer exponents cause no problem. The coefficient norm is Euclidean in the specified monomial basis. The data norm is exactly the Euclidean norm of (15), not an unstated ordinary derivative norm.

Theorem 5.1 (Exact moment-block spectrum)**.** *After a unitary discrete Fourier transform across G for each derivative order, followed by row and column permutations, $\mathcal{J}$ is the direct sum of the $h \times n_\eta$ matrices*

$$V_\eta = (d_\nu e_1^{\nu_1} e_2^{\nu_2})_{|\nu|<m,\ e\in\Omega_\eta}, \qquad h = \binom{m+1}{2}. \tag{16}$$

Empty column blocks may be ignored. Consequently,

$$\operatorname{rank} \mathcal{J} = \sum_\eta \operatorname{rank} V_\eta, \tag{17}$$

$$\sigma_{\min}(\mathcal{J}) = \min_{n_\eta > 0} \sigma_{\min}(V_\eta), \tag{18}$$

$$(V_\eta^* V_\eta)_{e,f} = \sum_{|\nu|<m} d_\nu^2 (e_1 f_1)^{\nu_1} (e_2 f_2)^{\nu_2}. \tag{19}$$

Here $\sigma_{\min}$ is the minimum gain on the column space domain, zero for a noninjective matrix, including one with more columns than rows.

Proof. The characters $\chi_\eta|_G$ are all characters of the finite group G, and their normalized value vectors form an orthonormal basis. The Fourier matrix has entries $r^{-1/2}\overline{\chi_\eta(g)}$. Multiplication with (15) yields

$$d_\nu e_1^{\nu_1} e_2^{\nu_2} \frac{1}{r} \sum_{g\in G} \chi_{e-\eta}(g) = d_\nu e_1^{\nu_1} e_2^{\nu_2}\, 1_{e\in\eta+M}.$$

Orthogonality can also be proved by translating the sum by a group element on which a nontrivial character is not one. The block form follows. Unitary transformations preserve the Euclidean Gram spectrum, and the Gram of the direct sum is the direct sum of the Grams. Taking the minimum Rayleigh quotient proves (18). Direct multiplication gives (19). □

The source already uses exact character projection to isolate coset components. The additional statement is an explicitly normalized conditioning identity and its error consequences, not a new character orthogonality principle. With rational d_ν, all block entries and Gram entries are rational, even when point coordinates are nonrational roots of unity. Positive definiteness can be certified by exact elimination or rational Cholesky/LDL data; decimal singular values alone are not a proof.

5.1 Coordinate and frame costs cannot be dropped

For one variable,

$$(X\partial_X)^a = \sum_{j=0}^{a} S(a,j) X^j \partial_X^j, \tag{20}$$

where $S(a,j)$ are Stirling numbers of the second kind. Applying the recursion $S(a+1,j) = jS(a,j) + S(a,j-1)$ proves this by induction. In two variables take the product of the two formulas. At each $g \in G$ this is a triangular invertible change from Taylor-normalized jets of total order below m to Euler jets of the same orders; its diagonal contains $a!b!g_x^a g_z^b$ and the specified positive row weights. Thus $\mathcal{J} = TJ_{\text{Taylor}}$ for a known invertible row matrix T including the $r^{-1/2}$ normalization. In Euclidean norms,

$$\sigma_{\min}(J_{\text{Taylor}}) \geq \sigma_{\min}(\mathcal{J}) / \|T\|_2 . \tag{21}$$

Neither T nor its inverse is generally unitary. A monomial coordinate map, a nonvanishing frame multiplier, or multiplication by a Laurent monomial likewise acts by an invertible finite jet matrix, not usually an isometry. A frame multiplier u acts by Taylor convolution with the coefficients of u, with diagonal $u(g) \neq 0$. Every subsequent margin must either retain these matrices or declare the transported norm.

5.2 Blockwise error and leakage

Let $V = \bigoplus_\eta V_\eta$ be full column rank, with specified left inverses $L_\eta V_\eta = I_{n_\eta}$. Write a perturbation in the same Fourier row coordinates as blocks $E_{\eta\theta}$ of size $h \times n_\theta$. Components in empty row-character blocks may be ignored by this left inverse, not by an assertion that they vanish.

Proposition 5.2 (A leakage-aware certificate)**.** *Set $b_{\eta\theta} = \|L_\eta E_{\eta\theta}\|_2$, bounding these quantities above rigorously if necessary. If positive z_η and $0 \leq \kappa < 1$ satisfy*

$$\sum_\theta b_{\eta\theta} z_\theta \leq \kappa z_\eta \quad \textit{for every nonempty column block } \eta, \tag{22}$$

then $V + E$ is injective. A left inverse is $(I + LE)^{-1}L$, $L = \bigoplus L_\eta$, with inverse-factor norm at most $(1-\kappa)^{-1}$ in the block norm $\max_\eta \|c_\eta\|_2 / z_\eta$.

Proof. The triangle inequality bounds the ηth component of LEc by $\sum_\theta b_{\eta\theta} \|c_\theta\|_2$. Divide by z_η and use (22). The Neumann argument from Theorem 4.2 applies. Errors in approximate identities $L_\eta V_\eta = I$ must be added to the diagonal bound. □

For character-preserving errors only the diagonal blocks matter: the test is $\max_\eta b_{\eta\eta} < 1$, with no artificial sum over r characters. Moving the points or passing through the source's analytic compression generally creates off-diagonal leakage. It must be bounded, not treated as symmetry-preserving just because the limiting test is. The anisotropic expansion expresses that leakage as known monomial directions; (12) or blockwise versions can retain it exactly.

5.3 A small exact subgroup test

Take the purely combinatorial triangle $P = \text{conv}\{(0,0),(2,0),(0,4)\}$, of area $4 = H/2$ with $H = 8$, $r = 2$, $k = 1$, and $m = 3 > \sqrt{H/r} = 2$. For $M = \mathbb{Z}(1,0) + \mathbb{Z}(0,2)$, $G = \{(1,1),(1,-1)\}$. There are nine exponent columns in $P \cap \mathbb{Z}^2$, and twelve Euler rows. Use all $d_\nu = 1$ and order the six

derivative orders as $(0,0),(0,1),(0,2),(1,0),(1,1),(2,0)$. The even-second-coordinate block has columns $(0,0),(0,2),(0,4),(1,0),(1,2),(2,0)$ and matrix

$$V_0 = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 2 & 4 & 0 & 2 & 0 \\ 0 & 4 & 16 & 0 & 4 & 0 \\ 0 & 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 & 1 & 4 \end{pmatrix}. \tag{23}$$

Exact elimination gives $\det V_0 = 64$ and $\left\|V_0^{-1}\right\|_\infty = 35/8$. The odd block has columns $(0,1),(0,3),(1,1)$. Its rows $(0,0),(0,1),(1,0)$ give determinant 2, so it too is injective. This independently executable example tests the stated source-subgroup mechanism. The artificial symbol $H = 8$ here denotes twice the triangle area; no polarized surface or nodal divisor realizing these data is constructed.

For a general finite exponent set, directly assembling (16) needs hN entries rather than rhN entries of the full point matrix. Block Gram storage is $\sum_\eta n_\eta^2$ rather than N^2 entries, and a straightforward Gram calculation takes $O(h\sum_\eta n_\eta^2)$ scalar operations. This excludes exponent enumeration, lattice construction, coefficient bit growth, certified analytic bounds, and dense leakage. It is not a uniform complexity bound for the geometric theorem.

6 Quantitative composition and the all-order boundary

6.1 Two scales, chosen in the correct order

The compression scale σ and the germ-dilation scale s solve different finite problems. The former can also be bounded explicitly. For a compression column with integers q, β, h, on a compact torus neighborhood with $0 < U_- \le |u| \le U_+$, $|y| \le Y$, let $U_q = \max(U_-^q, U_+^q)$. For $|\sigma|Y < 1$, the elementary identity

$$(1+\sigma y)^\beta - 1 = \beta\sigma y \int_0^1 (1+t\sigma y)^{\beta-1}\, dt$$

gives the valid bound, for positive or negative integer β,

$$\left|u^q y^h\big((1+\sigma y)^\beta - 1\big)\right| \le U_q Y^h\, |\beta|\, |\sigma| Y (1-|\sigma|Y)^{-(|\beta|+1)}. \tag{24}$$

For $\beta \ge 1$, use $(1+v)^{\beta-1} \le (1-v)^{-(|\beta|+1)}$, $0 \le v < 1$; for $\beta \le 0$ use the distance $|1+t\sigma y| \ge 1-v$. The case $\beta = 0$ is exact zero. Cauchy estimates on larger compact neighborhoods give the corresponding normalized derivative bounds, with the same margin factors as (8).

For each fixed (k,m) the resulting pipeline is precise:

1. Start with an injective triangle test at G, supplied by the source's geometric/lattice construction. Certify its blocks in the Euler metric, then retain the row-change costs for the required Taylor metric.

2. Restrict columns to the compressed slice basis. Use (24) and a left inverse to choose $\sigma \ne 0$ so that compression remains injective. Pull the points back by the actual invertible map $(u,y) \mapsto (u,(1+\sigma y)/u)$.

3. The resulting Q_t monomial test is injective. Restrict to the actual least-exponent set $E_t(F)$ and form its left inverse. Apply Theorem 4.2, or the uncancelled version, to choose $s > 0$. The finite basis changes and dilation preserve exact injectivity of F.

Discarding columns of an injective matrix preserves injectivity, but the coefficient norm, basis changes, and claimed numerical margin must still be specified. The cancellation set is optional: failure of its rank condition does not invalidate the uncancelled source transfer. Certified output is one actual injective configuration for this finite pair, given the named exact and analytic input data.

6.2 Why finitely many certificates do not prove maximality

The source's subgroup theorem uses a primitive $u \in \mathbb{Z}^2$ whose determinant projection on P_t has width $\sqrt{rH}$. With a unimodular completion v_0 and lattice $M = \mathbb{Z}u + \mathbb{Z}(rv_0)$, each character component becomes, after a Laurent monomial shift, a polynomial supported in a triangle of area $k^2H/(2r)$ and vertical range $k\sqrt{H/r}$. The map $(X, Y) \mapsto (X, Y^r)$ is locally invertible at the identity, not globally one-to-one. The source's triangle estimate bounds the vanishing order by $k\sqrt{H/r}$, contradicting m above that value.

The underlying triangle argument is elementary but all-order: a triangle of area $\rho^2/2$ and vertical range ρ has a horizontal-slice length tent of height ρ. For a Laurent polynomial supported there, factor the largest common $(U-1)^e$ out of its nonzero horizontal rows. Those rows have width at least e, so their heights lie in an interval of length at most $\rho - e$. The remaining nonzero polynomial evaluated at $U = 1$ has Z-span at most $\rho - e$. Its order at $Z = 1$ is bounded by that span. Adding the factor e proves order at most ρ. The specialization is nonzero by the definition of e; omitting this point would invalidate the argument. Translations and Laurent units preserve orders. This is the source's valid local algebraic mechanism, not a new theorem credited to the report.

The source chooses r_0, then $t = t(r)$ and $G = G(r)$, *before* k, m. In its direction lemma the largest power of two $j \le \sqrt{rH}/(2dH)$ and an odd integer in a sufficiently long open interval give a primitive vector, while continuous variation of t gives the exact width. A finite matrix check cannot replace this simultaneous parameter construction. Nor does a small residual prove the polygon bound for every section of every $L^{\otimes k}$.

For clarity we reproduce the logical final implication, conditional on *all* homogeneous tests. Suppose a tuple satisfies

$$\ker\left(H^0(S, L^{\otimes k}) \longrightarrow \bigoplus_i \mathcal{O}_{S,p_i}/\mathfrak{m}_{p_i}^m\right) = 0 \quad \text{for all integers } k, m > 0 \text{ with } m > k\sqrt{H/r}, \tag{25}$$

where local frames are understood. Put $A = \pi^*L$, $E = \sum E_i$, $w = \sqrt{H/r}$, and $D_0 = A - wE$. If D_0 is not nef, an integral nonexceptional curve C has $D_0C < 0$. Small rational $\delta > 0$ gives $(D_0 - \delta C)^2 > 0$ and $A(D_0 - \delta C) > 0$. Then choose rational $v > w$ close enough that $D = A - vE - \delta C$ has those two properties. For large divisible k, $H^2(Y, \mathcal{O}(kD)) = 0$ because $A(K_Y - kD) < 0$ and A is nef. Surface Riemann–Roch gives $h^0(kD) \ge (k^2D^2 - kDK_Y)/2 + \chi(\mathcal{O}_Y) > 0$. Adding the effective divisor $k\delta C$ yields an effective $kA - kvE$, contrary to (25) with $m = kv > kw$. This handles unequal curve multiplicities without equating them. The surface duality, Riemann–Roch, and blow-up ideal identity used here remain standard geometric inputs; the source supplies their references.

Finally, for each (k, m) the failure of full column rank of the algebraic principal-parts evaluation map is closed, defined by maximal minors. An analytic witness gives a nonempty Zariski-open injectivity locus. The countable intersection for fixed r is nonempty over $\mathbb{C}$ by the Baire argument on $U_r(\mathbb{C})$ and is precisely a very-general locus. Thus a different s and σ for every pair is allowed; a uniform analytic scale is not required. But a finite list of pairs, however well conditioned, is

not this countable all-order premise. The source's extension to uncountable algebraically closed characteristic-zero fields also requires finite-data descent and base change; none is certified by our numerical error bounds.

7 Evidence, limitations, and research directions

7.1 Literal reading and corrections

The owned read ledger records each public and private file's pre/post SHA256, total lines, literal intervals, and unread complements. Public scientific reads include all of `jets.tex` (198 lines), `node.tex` (193), `compression.tex` (157), `lattice.tex` (276), and `nef.tex` (287), as well as the 167-line introduction, figures, assembly files, README and bibliography. The cached line counts, not a different online rendering, determine this ledger. Snapshot authentication by the parent is separate evidence.

Fresh private-original reads total 462 lines: the finite-moment original [7], lines 155–233 and 455–526; the local-moment chapter [8], lines 1–166; and the stability original [9], lines 1–65 and 111–190. The chapter's two included supplement files were not opened. No complete private corpus, whole book, or every theorem cited by the source was read. The local proofs here use the actual finite moments, Taylor remainder, and preconditioned-error ideas, not a later comparative report as a replacement for originals.

No substantive correction to the source's finite jet lemmas was identified in this reading. Retained valid details are the strict gap after tie breaking, fixed finite derivative order, invertible nonunitary coordinate maps, exact character projections, nonzero specialized row polynomial, and the order of r, t, G versus k, m choices. The source does not claim that finitely many numerical ranks imply its global theorem; neither do we. The largest additional burden is obtaining certified analytic norms of actual section bases. These are input assumptions in our local results, not computed surface data supplied by the report.

7.2 Independent executable checks

The owned Python test file uses exact rational arithmetic and exact Gaussian-integer pairs, never a floating-point rank threshold. It checks positive rational-weight gaps on bounded exponent sets, Taylor/Euler change-of-row identities with negative Laurent exponents, character Gram blocks for roots of unity of orders two and four, the explicit subgroup minor and inverse norm, and the two-column cancellation and analytic response. It also checks rational compression inequalities and small leakage certificates. Counts and the test-script hash are recorded in `M096-checks.json`. The completed run passed 35,225 assertions across 17,394 bounded cases; these are author-run units, not parent counts. No actual surface section, nodal curve, maximal Seshadri constant, or global arithmetic/geometric theorem is tested by this finite suite.

7.3 Advantages and costs

Advantages. The gap is finite-search and truly positive, even for irrational positive weights. The left-inverse test can suppress large, precisely known low-order tail coefficients without changing the physical germ space. Character blocks avoid repeatedly representing roots of unity and give exact rational Gram data in the stated metric. Leakage budgets distinguish symmetry-preserving errors from errors that mix characters.

Costs. Cancellation has a real feasibility obstruction and can amplify noise. Large degrees can increase analytic majorants and inverse norms while shrinking admissible scales. Fourier unitarity belongs only to a specified Euler norm, not arbitrary frames or coordinates. Exact elimination can have large coefficient growth. None of these methods provides an effective geometric $r_0(S, L)$ from a small sample of tests.

Feasible extensions. Choose a finite cancellation shell by balancing its larger inverse norm against its improved Δ, retaining an exact rank test at every step. Optimize positive block weights in (22) using certified upper bounds rather than raw singular values. Develop interval majorants for explicitly presented local section bases and carry the actual frame matrices through both transfers. A future all-order numerical theorem would require bounds on these inputs as k, m vary and an independently proved geometric exhaustion; it is not obtained by extrapolating the finite examples.

7.4 Contribution assessment

The demonstrated increment is the combined, source-specific quantitative transfer: finite positive-weight tail bounds, exactly feasible annihilation of its low-weight defects, and Fourier moment-block conditioning with honest metric and leakage costs. Example 4.3 proves a strict gain in the admissible error budget. The classical quotient, Cauchy, Neumann, Stirling, and Fourier identities themselves are not credited as inventions. C3 records a notable local improvement; C4 or C5 would overstate the absence of new geometry, a computed surface family, or a new global theorem. L3 is unchanged.

References

[1] OpenAI, *Maximal Seshadri constants on arbitrary polarized surfaces*, September 23, 2026 release, cached source at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Public source directory.

[2] W. Syzdek and T. Szemberg, *Seshadri fibrations of algebraic surfaces*, Math. Nachr. 283 (2010), 902–908; arXiv:0709.2592, Conjecture 4.3.

[3] J. Alexander and A. Hirschowitz, *An asymptotic vanishing theorem for generic unions of multiple points*, Invent. Math. 140 (2000), 303–325; author abstract, version 3.

[4] C. Dionne and M. Roth, *Seshadri constants on $\mathbb{P}^1 \times \mathbb{P}^1$ and applications to the symplectic packing problem*, Forum Math. Sigma 13 (2025), e191, doi:10.1017/fms.2025.10137.

[5] A. Laface and L. Ugaglia, *Irrational Seshadri constants from dihedral orbits*, September 22, 2026 preprint, arXiv:2609.26521v1, Theorems 1–2.

[6] G. Malara, Ł. Merta, J. Szpond, and M. Zieliński, *Dihedral reflections and an infinite series of irrational Seshadri constants*, October 1, 2026 preprint, arXiv:2610.01783, abstract only inspected.

[7] M. Subbarao, confidential original local report `RamFoundationTRs/TR-rev-1-old-tr250/TR-rev-1-old-tr250.tex`, finite Ram representations and concrete polynomial construction; literal selected ranges in the owned read ledger.

[8] M. Subbarao, confidential original *Local Moments and Ram Differential Operators*, Volume 1 revision 2, chapter 3, `ch03-local-moments-rev2.tex`, lines 1–166.

[9] M. Subbarao, confidential original stability chapter, `RamNonlin-IE-PDE-astra/TR1-lin-nonlin/stability.tex`, finite perturbed inverse and preconditioned certificate proofs, lines 1–65 and 111–190.